```java
import java.util.Scanner;
import java.util.Arrays;

public class  Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int arr[] = new int[n];


        System.out.println("Enter the elements:");
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }


        System.out.println("\nChoose Search Method:");
        System.out.println("1. Linear Search");
        System.out.println("2. Binary Search");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();


        System.out.print("Enter value to search: ");
        int key = sc.nextInt();

        if(choice == 1) {


            boolean found = false;

            for(int i = 0; i < n; i++) {
                if(arr[i] == key) {
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                    System.out.println("Value found at index: " + i);
                    found = true;
                    break;
                }
            }

            if(!found) {
                System.out.println("Value not found.");
            }

        }
        else if(choice == 2) {


            Arrays.sort(arr);

            int low = 0;
            int high = n - 1;
            boolean found = false;

            while(low <= high) {
                int mid = (low + high) / 2;

                if(arr[mid] == key) {
                    System.out.println("Value found at index: " + mid);
                    found = true;
                    break;
                }
                else if(arr[mid] < key) {
                    low = mid + 1;
                }
                else {
                    high = mid - 1;
                }
            }

            if(!found) {
                System.out.println("Value not found."
```

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);
            }
        }
        else {
            System.out.println("Invalid choice.");
        }

        sc.close();
    }
}
```